Exchange in a Box: A Deterministic Laboratory for Market Microstructure

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Abstract

We describe a reproducible market-microstructure laboratory built around a pure limit-order-book transition function. The system includes price-time priority, multi-route replay, a checksummed journal, a static cross-venue router, account pretrade controls, a local control room, synthetic agents, and point-process diagnostics. We give invariants for matching and accounting, a routing optimality argument, exact likelihoods for one- and two-type Hawkes models, and a chronological held-out protocol. A paired synthetic intervention varies shock size and population regime under common random inputs. No result establishes empirical market fit, real-world causal impact, or trading profitability.

1 A transition system for an exchange

Let S_n denote the state after command n and c_n the next command. The engine is a function

$$F : S_n \times c_n \longrightarrow (S_{n+1}, o_n), \tag{1}$$

where o_n contains acceptance status and an ordered list of executions. State consists of bid and ask queues, a set of previously accepted IDs, and a monotone sequence counter. Prices $p \in \mathbb{Z}_{>0}$ are ticks and quantities $q \in \mathbb{Z}_{>0}$ are lots. The current command language contains submissions and cancellations. Matching executes against the best opposite quote, using the earliest accepted resting order at a tied price. Execution price is the resting order's limit price.

For a buy taker with limit p, a resting ask at a is eligible exactly when $a \le p$. For a sell taker, a resting bid at b is eligible exactly when $b \ge p$. The matched quantity is $\min(q_{\text{taker}}, q_{\text{maker}})$, and matching repeats until the taker is exhausted or no eligible maker remains. Residual taker quantity enters its side of the book. Rejected commands leave state unchanged.

Proposition 1 (Quantity conservation for an accepted submission)**.** *For an accepted submission with initial quantity q, let $x_1, \dots, x_k$ be its execution quantities and r its residual resting quantity, taking $r = 0$ when the order is fully executed. Then*

$$q = \sum_{i=1}^{k} x_i + r. \tag{2}$$

Proof. Each matching step subtracts the same positive amount x_i from the taker's remaining quantity as it emits in an execution. The loop terminates with the remaining quantity r, possibly zero. Telescoping these subtractions gives the claim. □

Proposition 2 (Uncrossed resting book)**.** *In every reachable state with both sides nonempty, the best bid is below the best ask.*

Proof. The empty book satisfies the claim. Cancellation cannot create a crossing. On submission, an incoming order consumes every eligible opposing resting order until its quantity is zero or the best opposing price is outside its limit. Only in the latter case can a residual order rest, and its limit is then strictly outside the best opposing price. Induction over commands completes the proof. □

Pure transitions give a direct replay property: for any fixed command sequence C, folding F from the same initial state yields the same final state and outputs. A route key (v, m) indexes venue v and instrument m; an event changes only its selected book. The scheduler sorts by delivery time and then original input position, making latency and drops explicit deterministic inputs. Global event timestamps are nondecreasing after scheduling.

The versioned journal stores route, timestamp, command, sequence number, and a checksum of each complete record. Reopening validates the complete prefix and replays it. An incomplete final record can be ignored during recovery and removed by an explicit repair step; a complete corrupt record is rejected. The checksum detects accidental corruption, not adversarial tampering. Buffered single-writer append improves long-run cost, but flush and close do not provide an fsync power-loss guarantee. Account risk requests and ledger state are not journaled.

2 Static cross-venue routing

For a single instrument, let venue v display a price-time ordered sequence of makers j with available quantity q_{vj} and price p_{vj}. Given a target buy quantity Q, fee f_v, and chosen latency penalty d_v, the planner minimizes

$$\sum_{v,j}(p_{vj} + f_v + d_v)x_{vj}, \qquad \sum_{v,j} x_{vj} = \min\left(Q, \sum_{v,j} q_{vj}\right), \quad 0 \le x_{vj} \le q_{vj}, \tag{3}$$

subject to consuming each venue's queue in order. For a sell, replace p_{vj} in the unit cost by $-p_{vj}$. A min-priority queue exposes the next maker from each venue and repeatedly takes the lowest available unit cost. Constant venue fees and penalties make unit costs nondecreasing along a venue's price-time queue.

Proposition 3 (Static allocation optimality)**.** *Under the displayed-depth and constant-cost assumptions above, the min-priority-queue allocation has minimum total cost among maximum-fill allocations that preserve every venue's maker order.*

Proof. At each step, the first unconsumed maker of every venue is the only currently eligible maker from that venue. Any feasible allocation that uses a higher-cost eligible unit while leaving a lower-cost eligible unit unfilled can swap those units without reducing fill or breaking a queue prefix. Repeating the exchange yields the greedy allocation. Equal-cost ties use venue name and maker sequence only for determinism. □

This is a static plan, not an order submission. Displayed depth may disappear, and the latency penalty is an objective weight rather than measured execution cost. Competing flow, partial-fill uncertainty, hidden depth, and venue-specific lot rules are not modeled.

3 Pretrade controls and accounting

Each synthetic account has inventory P_a, cash C_a, resting buy quantity R_a^+, resting sell quantity R_a^-, and configured limits. For an incoming order with signed quantity σq, where $\sigma = 1$ for a buy and -1 for a sell, a conservative directional inventory check is

$$|P_a + \sigma q + \sigma R_a^{\sigma}| \le L_a^{\text{position}}, \qquad pq \le L_a^{\text{notional}}, \qquad R_a^+ + R_a^- + q \le L_a^{\text{open}}. \tag{4}$$

The implementation rejects self-trades and unauthorized cancellations. It supports four atomic policies: good till cancelled, immediate or cancel, fill or kill, and post only. A rejected request leaves the prior state intact. Halting an account cancels its resting orders and prevents new placements until resumed.

The limits count submitted quantity before immediate execution. They can therefore reject an order that a less conservative policy would admit. Cash is an accounting balance, not a collateral or margin model.

Proposition 4 (Paired-fill ledger conservation)**.** *If every maker and taker belongs to a tracked account, each execution preserves aggregate inventory and aggregate cash:*

$$\sum_a \Delta P_a = 0, \qquad \sum_a \Delta C_a = 0. \tag{5}$$

Proof. For an execution of x lots at price p, the buy account receives $(+x, -px)$ and the sell account receives $(-x, +px)$. Summing each coordinate cancels. A rejected request creates no execution. □

4 Order-book state as a mathematical object

Let B and A be the best bid and ask prices, with displayed quantities Q_B and Q_A aggregated across all orders at those prices. When both sides are present, define spread, imbalance, and microprice by

$$s = A - B, \tag{6}$$

$$I = \frac{Q_B - Q_A}{Q_B + Q_A}, \tag{7}$$

$$M = \frac{AQ_B + BQ_A}{Q_B + Q_A}. \tag{8}$$

Then $I \in [-1, 1]$ and $B \le M \le A$. These are state summaries, not forecasts. Predictive use would require out-of-sample evaluation with exchange-specific data, costs, queue position, and regime analysis.

5 Stochastic arrival models

As a candidate model for simulated arrivals, consider the univariate exponential Hawkes intensity [1]

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} e^{-\beta(t - t_i)}, \qquad \mu > 0, \quad \alpha \ge 0, \quad \beta > 0, \quad \alpha/\beta < 1. \tag{9}$$

The final inequality is the usual stationary branching condition. Our paths have empty prehistory at time zero; subcriticality does not imply stationary initialization. For ordered event times $0 \le t_1 < \cdots < t_N \le T$, the exact log likelihood under this observation convention is

$$\ell(\mu, \alpha, \beta) = \sum_{i=1}^{N} \log \lambda(t_i) - \mu T - \frac{\alpha}{\beta} \sum_{i=1}^{N} \left(1 - e^{-\beta(T - t_i)}\right). \tag{10}$$

The implementation evaluates the pre-event intensity by a decaying recurrence, so likelihood evaluation is linear in N rather than quadratic. It also computes compensator increments

$$z_i = \int_{t_{i-1}}^{t_i} \lambda(u)\, du, \qquad t_0 = 0, \tag{11}$$

which support time-rescaling diagnostics [3]. If the conditional intensity is correctly specified and its parameters are known, the increments have independent unit-exponential distributions. Estimation and finite windows complicate a formal goodness-of-fit test, so we report descriptive residual summaries only.

Bounded, fixed-seed thinning simulation follows the conditional-intensity method of [2]. Exhaustion of an event or candidate budget is an error rather than a silently truncated path.

For marked buy/sell events, let $r, s \in \{b, a\}$ index event types and let $A_{rs} \geq 0$ be the excitation of target r by source s. The two-type candidate model is

$$\lambda_r(t) = \mu_r + \sum_{s \in \{b,a\}} A_{rs} \sum_{i: t_i < t,\, m_i = s} e^{-\beta(t - t_i)}, \qquad \rho(A/\beta) < 1, \tag{12}$$

where ρ is spectral radius. Its exact marked log likelihood on $[0, T]$ is

$$\ell = \sum_{i=1}^{N} \log \lambda_{m_i}(t_i) - T \sum_r \mu_r - \frac{1}{\beta} \sum_{i=1}^{N} \sum_r A_{r,m_i} \left(1 - e^{-\beta(T - t_i)}\right). \tag{13}$$

The implementation computes component and aggregate compensator increments and bounded seeded simulations from empty prehistory. No multivariate fit or empirical calibration is claimed.

6 Chronological evaluation and execution metrics

For a split $\tau < T$, estimation uses events in $[0, \tau]$ only. Held-out score is the conditional likelihood

$$\ell_{(\tau,T]}(\theta) = \sum_{i: t_i > \tau} \log \lambda_\theta(t_i \mid \mathcal{H}_{t_i^-}) - \int_\tau^T \lambda_\theta(u \mid \mathcal{H}_{u^-})\, du. \tag{14}$$

The likelihood retains pre-split history inside the intensity while integrating only the held-out interval. Poisson uses its training-count rate; the univariate Hawkes comparator selects the best training likelihood from a declared finite grid. This is grid selection, not continuous maximum-likelihood estimation. The first compensator interval crossing the split is omitted from held-out residual summaries.

For a taker side $\sigma \in \{+1, -1\}$, execution VWAP V, arrival midpoint M_0, and later midpoint M_1, define effective spread $E = 2\sigma(V - M_0)$, realized spread $R = 2\sigma(V - M_1)$, and signed midpoint change $J = \sigma(M_1 - M_0)$. Then $E = R + 2J$ identically. These are descriptive decompositions; J alone is not a causal market-impact estimate.

7 A paired intervention inside the simulator

For random seed s, population regime r, and extra buy size q, run a control and shocked book in lockstep. Every synthetic population innovation $U_{s,t}$ is drawn once and applied to both arms. Arm-specific quotes may then produce different prices and future states. With final midpoints $M_{s,T}^{(q,r)}$ and $M_{s,T}^{(0,r)}$, the predeclared paired outcome is

$$D_s(q, r) = M_{s,T}^{(q,r)} - M_{s,T}^{(0,r)}. \tag{15}$$

The saved protocol fixes forty seeds, 1,000 population events per arm, shock sizes 0, 200, and 800 lots, and passive and active agent regimes. It also records shock execution cost relative to the initial midpoint. A percentile bootstrap resamples complete seed pairs 5,000 times to describe uncertainty of the mean conditional on this simulator. Table 1 reports the four nonzero-shock configurations. All final books had two-sided quotes, and zero-shock arms were identical.

The 200-lot and 800-lot shocks fill against the fixed initial ladder at respective costs of 300 and 3,600 tick-lots. The paired contrast has a defined counterfactual inside the simulator, but its response is conditional on the declared agent rules. It is not an empirical causal-impact estimate.

Regime	Shock lots	Mean D_s (ticks)	Bootstrap 95% interval
Passive	200	1.475	[1.325, 1.625]
Passive	800	3.500	[3.025, 4.025]
Active	200	1.1625	[1.025, 1.300]
Active	800	3.950	[3.4625, 4.425]

Table 1: Paired midpoint differences in a fixed synthetic population. Intervals describe variation over the chosen seeds and bootstrap procedure, not uncertainty about real markets.

8 Synthetic checks and limits

Table 2 summarizes distinct verification exercises. Counts are not exchange-data sample sizes. The saved arrival-model experiment uses ten fixed-seed Poisson paths and ten fixed-seed Hawkes paths, each near 40,000 events over a horizon of 10,000 time units. The generating Hawkes branching ratio and decay lie on the candidate grid; baseline candidates depend on the training count. All ten Poisson paths select zero excitation; all ten Hawkes paths select branching ratio 0.5 and decay 1.2. This favorable recovery check says little about model selection under misspecification or changing regimes.

Synthetic verification exercise	Scale
Multi-route matching and replay	120,000 events, 12 seeds
Static router oracle and depth stress	500 oracle cases; 50,000 makers
Persistent journal reopen, repair and replay	60,001 records
Account and policy randomized checks	50,000 requests
Population scenario deterministic reruns	480,000 events
Univariate arrival-model comparisons	20 paths, about 800,000 events
Two-type aggregate compensator equivalence	399,455 events, 40 seeds
Paired synthetic interventions	240 pairs, 480,000 arm events
Separate Poisson likelihood stress path	999,409 events

Table 2: Software and synthetic-model checks. Runs exercise different paths and must not be combined into one empirical sample.

The saved study is fully synthetic and uses the same grid construction on every path. It does not test calibration to real markets, a causal intervention, profitability, or a strategy. A later study would need data rights and venue rules, timestamp resolution, queue position, fees, parameter sensitivity, non-stationarity, leakage controls, and regime dependence. Journal durability, concurrent writers, and realistic broker/exchange connectivity are outside the present implementation.

9 Conclusion

A deterministic matching core, replayable journal, and explicit point-process models make synthetic market-mechanism experiments auditable. The present artifact is a tested software laboratory. Its research claims remain deliberately narrow until empirical data and prospectively specified evaluations are available.

References

[1] A. G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971. doi:10.1093/biomet/58.1.83.

[2] Y. Ogata. On Lewis' simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1):23–31, 1981. doi:10.1109/TIT.1981.1056305.

[3] E. N. Brown, R. Barbieri, V. Ventura, R. E. Kass, and L. M. Frank. The time-rescaling theorem and its application to neural spike train data analysis. *Neural Computation*, 14(2):325–346, 2002. doi:10.1162/08997660252741149.